

AI Agents and Agentic AI Comprehensive Beginner Handout

DATA SCIENCE COURSE — BSIT PROGRAM

Concepts • Workflows • Tools • GitHub Copilot • Responsible Use

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1 Introduction to AI Agents

AI Agent

An AI agent is a system that can **understand a goal, decide what to do next, use available tools, observe results, and continue until the task is finished or stopped**. It is more than a simple chatbot because it can work through **multiple steps** instead of only giving one direct answer.

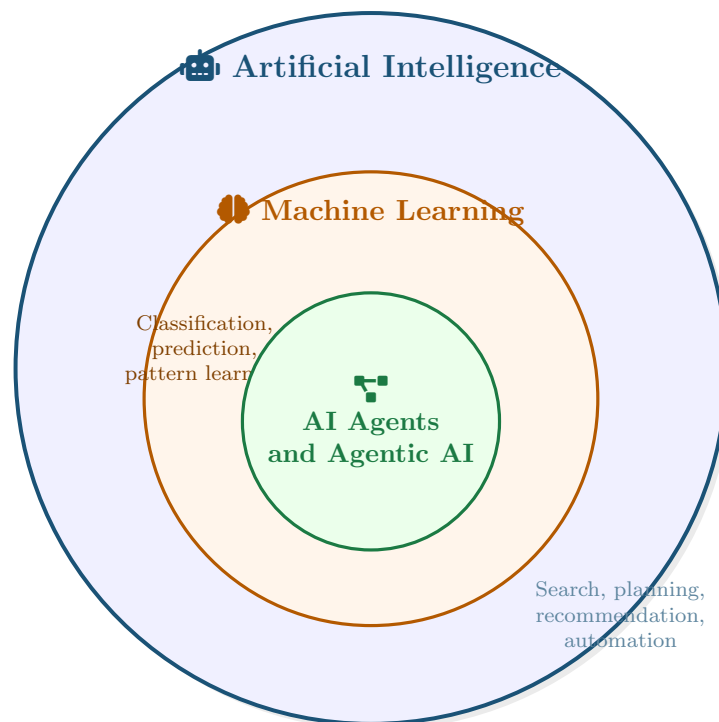
Agentic AI

Agentic AI refers to AI systems that behave in a more goal-directed and action-oriented way. These systems can **plan, reason, use tools, keep track of state, and take follow-up actions** with limited human intervention.

Simple Meaning

If a system only answers one question at a time, it behaves like a basic assistant. If it can break a task into steps, search for information, write files, call tools, and revise its work, it is acting in a more **agentic** way.

1.1 Where AI Agents Fit in AI



- **Artificial Intelligence** is the broad field of making systems act intelligently.
- **Machine Learning** is a subset that learns from data.
- **AI Agents** are intelligent systems that pursue goals through steps, decisions, and actions.

1.2 Why AI Agents Matter

- They help automate repetitive and multi-step work.

- They can combine reasoning with tools such as search, files, APIs, and databases.
- They are useful in education, business, software engineering, data analysis, and customer support.
- They represent a practical shift from simple AI responses to task completion.

2 AI Agent vs Chatbot vs Traditional Automation

System	Main Behavior	Strength	Limitation
Basic chatbot	Answers direct questions	Easy to use	Usually limited to one-turn responses
Traditional automation	Follows fixed rules or scripts	Reliable for repeated tasks	Cannot adapt well to new situations
AI agent	Plans, acts, checks results, and continues	More flexible and capable	Needs guardrails and monitoring

💡 Key Difference

A chatbot mainly **responds**. An AI agent can **respond, decide, act, observe, and continue working**.

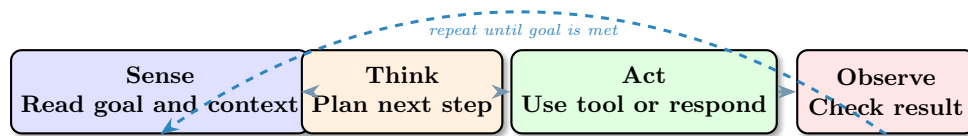
3 Core Components of an AI Agent

Component	Purpose
Goal or instruction	Defines what the agent is trying to achieve
Model or reasoning engine	Interprets input and decides what to do next
Memory or state	Stores conversation history, facts, or intermediate work
Tools	Allow the agent to search, calculate, code, call APIs, or edit files
Environment	The world or system the agent interacts with
Guardrails	Safety rules, permissions, and limits on what the agent can do
Feedback loop	Lets the agent check progress and revise if needed

3.1 Easy Interpretation of the Components

- The **goal** tells the agent what success looks like.
- The **model** provides reasoning and generation ability.
- **Memory** prevents the agent from starting from zero every time.
- **Tools** let the agent do real work beyond text generation.
- **Guardrails** reduce misuse, errors, and unsafe actions.

4 The Agent Loop: Sense, Think, Act, Observe



Agent Loop

Most AI agents operate through a repeating cycle:

- **Sense:** Understand the prompt, files, history, or environment.
- **Think:** Decide what should happen next.
- **Act:** Use a tool, generate content, or modify something.
- **Observe:** Review the result and decide whether to continue.

5 Memory, Tools, and Planning

5.1 Memory

Memory in Agents

Memory allows an agent to store useful information from previous steps, such as user preferences, earlier outputs, temporary notes, or task history.

- **Short-term memory:** Current conversation or current task state.
- **Long-term memory:** Stored preferences, facts, or reusable knowledge.

5.2 Tools

Tool Use

Tool use means the agent can call external capabilities rather than only generate text. Tools may include web search, file editing, database access, Python execution, calculators, or application commands.

</> Simple Example

If a user asks, “*Create a report from this CSV file,*” a basic chatbot may only describe the steps. An agentic system can open the file, analyze the data, generate the report, and save the result.

5.3 Planning

💡 Planning

Planning means breaking one large task into smaller steps. For example, instead of trying to solve everything at once, an agent can first inspect the data, then clean it, then analyze it, then summarize the results.

6 Types of AI Agents

Type	Main Idea	Example
Single agent	One agent handles the whole task	A coding assistant that completes one workflow
Multi-agent system	Several agents share responsibilities	Planner agent plus researcher agent plus reviewer agent
Reactive agent	Responds based on current situation	A support bot answering FAQ prompts
Goal-based agent	Chooses steps to reach a target	An assistant that prepares a complete project file
Tool-using agent	Calls external tools to act	A data agent that runs SQL or Python

6.1 Single-Agent vs Multi-Agent

Single-Agent System	Multi-Agent System
Simpler to build and manage	More modular and specialized
Fewer coordination problems	Better for dividing complex tasks
Easier to monitor	More overhead and more communication complexity
Useful for focused tasks	Useful for large workflows or teams of functions

7 Large Language Models and AI Agents

📖 LLM as the Reasoning Core

Modern AI agents are often built around Large Language Models (LLMs). The LLM helps the agent understand instructions, generate explanations, choose next steps, and interact in natural language.

💡 Important Distinction

An LLM by itself is not always a full agent. It becomes part of an agentic system when it is connected to **goals, memory, tools, and feedback loops**.

- An LLM helps with language understanding and generation.
- Tools help the system take external actions.
- Memory helps it keep track of what has happened.
- Planning helps it complete multi-step tasks.

8 Key Beginner Terms

Term	Meaning
Prompt	The instruction given to the system
State	The current situation or memory of the task
Action	Something the agent does, such as calling a tool or writing text
Observation	The result of an action
Tool call	A request sent to a calculator, file system, API, or application
Planning	Dividing a task into steps
Reasoning	Deciding what action makes sense next
Guardrail	A control that limits unsafe or incorrect behavior
Autonomy	The amount of work the agent can do without direct human guidance
Human in the loop	A human reviewer approves or checks important steps

9 GitHub Copilot as an Example of Agentic AI

GitHub Copilot

GitHub Copilot is an AI coding assistant that helps users write, explain, edit, and navigate code. In modern development environments, it is not limited to autocomplete. It can also support chat-based help, multi-file edits, and more agentic workflows.

9.1 Why Copilot Matters in Data Science

- It can help write Python, SQL, R, and notebook code.
- It can explain libraries, functions, and error messages.
- It can help generate visualizations, summaries, and documentation.
- It can support experimentation, debugging, and project organization.

9.2 Common Copilot Experiences

- **Inline code completion:** Suggests code while you type.
- **Chat assistance:** Answers questions and explains code or concepts.
- **Editing assistance:** Applies requested changes to code or text.
- **Agentic workflow support:** Can inspect a workspace, make a series of edits, and validate changes.

9.3 GitHub Copilot Modes

💡 Mode Names Can Vary by Version or Editor

GitHub Copilot features change over time. In VS Code and similar environments, the most important beginner-level distinction is between **asking**, **editing**, and **agent-like task execution**. The names and exact buttons may vary across versions.

Mode	What It Does	Typical Use
Ask mode	Answers questions, explains code, suggests approaches	Learn concepts, debug errors, understand algorithms, ask for examples
Edit mode	Applies requested changes to code or text	Refactor functions, update notebooks, fix formatting, rewrite blocks
Agent mode	Works through a multi-step task using repo context and tools	Search files, update several places, run validations, complete coding tasks

9.4 How Students Should Use Copilot

- Use **Ask mode** to understand topics before copying code.
- Use **Edit mode** when you already know what you want changed.
- Use **Agent mode** for larger tasks that involve multiple steps or files.
- Always test and review outputs, especially for assignments or data analysis.

⚠️ Important Academic Note

Copilot can speed up learning, but students should not rely on it blindly. The user is still responsible for correctness, originality, and understanding the submitted work.

10 Applications of AI Agents in Data Science

- **Data collection assistance:** gathering information from files, APIs, or structured sources.
- **Data cleaning support:** suggesting preprocessing steps and detecting common issues.
- **Exploratory data analysis:** generating charts, summaries, and descriptive interpretations.
- **Modeling assistance:** recommending algorithms, hyperparameters, and evaluation approaches.
- **Documentation and reporting:** summarizing findings into readable reports.
- **Notebook guidance:** helping explain code cells, functions, and outputs.

10.1 BSIT-Relevant Examples

Examples for BSIT Students

- An agent that explains Python code in a Jupyter notebook.
- An agent that cleans CSV data and produces a short summary.
- An agent that drafts documentation for a class project.
- An agent that helps compare machine learning models and metrics.
- A campus support bot that answers student questions using official information.

11 Benefits of AI Agents

- Reduce repetitive work.
- Combine information from many tools and sources.
- Support faster workflows and decision-making.
- Provide more interactive and personalized assistance.
- Help users move from ideas to results more quickly.

12 Risks, Limitations, and Challenges

Risk	Meaning	Possible Response
Hallucination	The system produces false or unsupported information	Verify important outputs and use trusted data
Unsafe tool use	A tool call may change or expose data incorrectly	Add permissions and approval checkpoints
Prompt injection	Malicious instructions try to manipulate the system	Validate inputs and restrict tool access
Privacy risk	Sensitive data may be revealed or misused	Avoid unnecessary sharing and protect confidential data
Over-automation	Users trust the system too much	Keep humans involved in critical decisions

Hallucination

Hallucination means the AI gives an answer that sounds correct but is actually false, incomplete, or invented.

Prompt Injection

Prompt injection is an attack or manipulation attempt where the input tries to override the intended rules of the system and force it into unsafe or incorrect behavior.

13 Best Practices for Building or Using Agents

- Start with a narrow and clear task.
- Limit what tools the agent can use.
- Keep logs of important actions.
- Validate important outputs before using them.
- Use human approval for high-impact steps.
- Measure accuracy, latency, cost, and safety.
- Test with normal inputs and edge cases.
- Explain to users what the agent can and cannot do.

💡 Practical Rule

The best beginner agent systems are not the most autonomous ones. They are the ones that are **clear, narrow, testable, and safe**.

14 Mini Case Study for BSIT Data Science

📄 Scenario

A BSIT Data Science class wants an assistant that can help students understand assignments, explain dataset columns, generate starter analysis code, and summarize notebook results.

Possible workflow:

1. The student asks a question about a dataset or notebook.
2. The agent reads the prompt and relevant files.
3. It plans what information to inspect first.
4. It generates an explanation or starter code.
5. It checks whether the result matches the student's requested level.
6. The student reviews and tests the answer before using it.

⚠️ Responsible Use in Education

An educational agent should support learning, not replace understanding. Students still need to read, test, and explain their own work.

15 A Small Pseudocode Example

```
goal = "Summarize this dataset and suggest first analysis steps"

context = read_file("students.csv")
plan = agent.plan(goal, context)
result = agent.use_tools(plan)
checked = agent.review(result)

print(checked)
```

 What This Shows

The agent receives a goal, reads context, creates a plan, uses tools, reviews the result, and then returns an answer. This is the basic logic behind many modern agent systems.

16 Summary

- AI agents are systems that pursue goals through repeated cycles of understanding, acting, and checking.
- Agentic AI combines reasoning, planning, memory, and tool use.
- Agents differ from simple chatbots because they can complete multi-step tasks.
- Modern agents often use LLMs, but LLMs alone are not full agents.
- GitHub Copilot is a useful example of agentic assistance in coding and Data Science workflows.
- Copilot is commonly understood through three practical modes: Ask, Edit, and Agent.
- AI agents can be powerful, but they require validation, guardrails, and responsible use.

17 Review Questions

1. What is the difference between an AI agent and a basic chatbot?
2. What does the term agentic AI mean?
3. Why are memory and tools important in an agent system?
4. What happens in the sense-think-act-observe loop?
5. How does a single-agent system differ from a multi-agent system?
6. Why is an LLM not always a complete agent by itself?
7. What are the main purposes of GitHub Copilot Ask mode, Edit mode, and Agent mode?
8. What are some useful applications of AI agents in Data Science?
9. What is hallucination, and why is it dangerous?
10. Why should human review remain part of agent-based systems?